



Fig. 2: Fabrication and testing of author's prototype of the big-size automatic digital wheel counter

All these sub-sections/workstations are required to operate in a completely synchronised manner to produce forged wheels, which later undergo heat treatment and machining to produce finished wheels.

The challenges

Before taking up this project, there was no common display system for the entire wheel forge shop to view the production status (that is, digital counting display of forged wheels), which is required for proper feedback and control towards higher production. Only manual counting by the stamper or operator at other places was done, which made it difficult for employees and shop-floor managers to know the shift/daily forged output at any point of time.

During productivity improvement studies, it was felt that a modern digital counting and display system that is visible to all employees of the wheel forge shop would help get rid of this problem. This was necessary not only for the correct accounting of produced wheels but also for motivating employees towards higher production, primarily through feedback.

Hence, an innovation project was taken up to develop a digital wheel counter with a jumbo-sized LED display, without any considerable capital investment. Accordingly, a complete

three-digit electronic digital wheel counting system with a giant seven-segment display was successfully designed, fabricated and installed at Wheel and Axle Plant (WAP) Wheel Forge Shop, which is working well, and is giving several tangible and intangible benefits.

Salient features of the innovative three-digit electronic wheel counter are as follows:

- Display size is big enough (digit height is 45.7cm, or 1.5 feet) to be visible from all corners of the wheel forging and rolling section.
- The circuit and the entire automatic display system have been designed, developed and fabricated inhouse using waste materials and low-cost items.
- Display brightness and LED

matrix density is visible from far as well as from a close distance.

- Any reputed supplier/contractor is likely to charge an amount not less than 400,000 to 500,000 rupees for installing such an automatic wheel counting system with giant display. This innovative digital counter with giant LED display costs only 4500 rupees (approximately).

- The seven-segment display has been developed from common household items like plastic wiring casing, decorative LED strips and ply board, among others.

- The electronic digital counter circuit and giant LED display board have been developed using easily-available ICs and other electronic components.

- IC bases have been used for easy replacement of ICs in case of damage.

- Increment of wheel counter can be made by pushing the manual push-button switch as well as automatic limit switch, which gives a pulse during punching of the identification number on the hot wheel.

- There is an auto/manual selector switch for convenience, and a reset switch for resetting the digital counter to zero for shift or day start.

Benefits of the project

The innovative digital counter with three-digit jumbo display was successfully developed, fabricated and installed on the roof of the stamping area cubicle at the wheel forge area. Some of its benefits are given below.

- The counter display gives instantaneous production status of the wheel forge section, where time is directly proportional to wheel production numbers.

- The common display maintains production synchronisation among various workstations in the shop. It

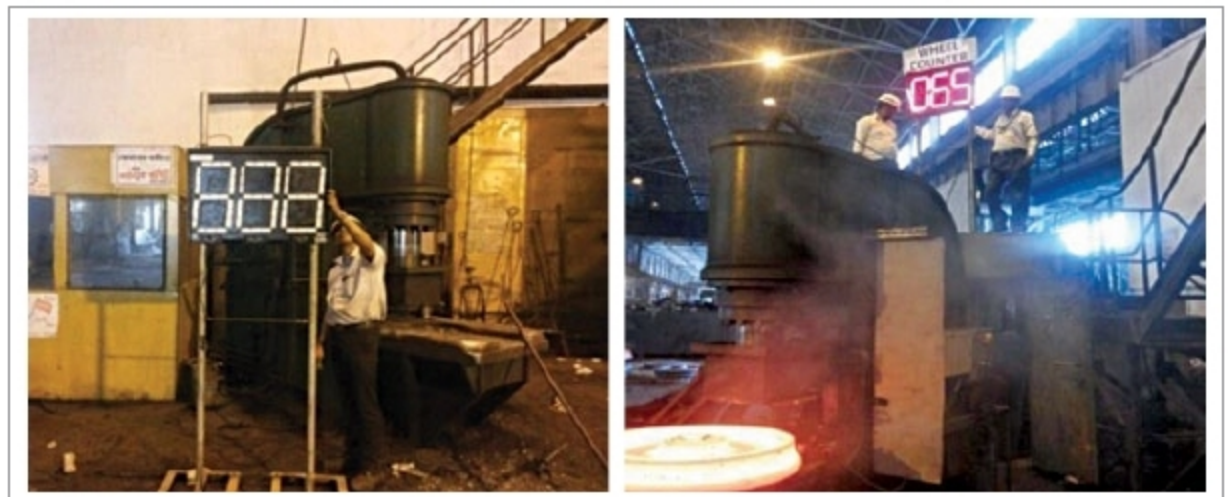


Fig. 3: Installation and testing of the jumbo seven-segment display above stamping press control cabin